

Ex. 2

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```
print(uuid::UUIDgenerate())
```

```
## [1] "3a9100d1-4d98-4cc1-8241-a065fa599e73"
```

Question 1

```
commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,  
            31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)
```

```
# 1. Range and Class Width
```

```
range_val <- max(commute) - min(commute)  
range_val
```

```
## [1] 27
```

```
class_width <- ceiling(range_val / 5)  
class_width
```

```
## [1] 6
```

```
# 2. Grouped Frequency Distribution using cut() and table()
```

```
breaks <- seq(17.5, 17.5 + 5 * class_width, by = class_width)  
classes <- cut(commute, breaks = breaks, right = TRUE)  
freq_table <- table(classes)  
freq_table
```

```
## classes
```

```
## (17.5,23.5] (23.5,29.5] (29.5,35.5] (35.5,41.5] (41.5,47.5]  
##           5           6           5           6           2
```

```
# 3. Cumulative Frequency using cumsum()
```

```
cum_freq <- cumsum(freq_table)  
cum_freq
```

```
## (17.5,23.5] (23.5,29.5] (29.5,35.5] (35.5,41.5] (41.5,47.5]  
##           5           11           16           22           24
```

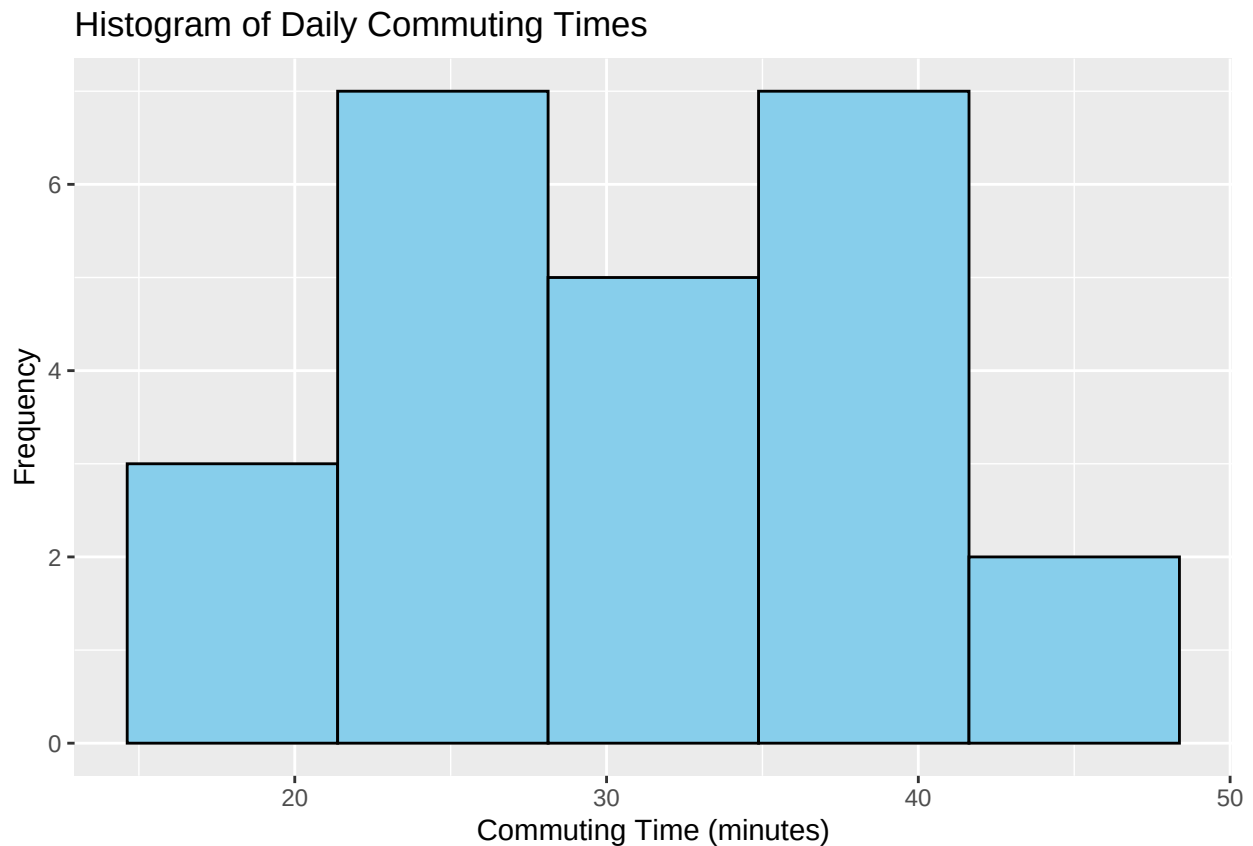
The range of the commuting times is 27 minutes (from 18 to 45). Dividing the range by 5 gives 5.4, which rounds up to a class width of 6.

Question 2

```
library(ggplot2)
```

```
ggplot(data.frame(commute), aes(x = commute)) +
```

```
geom_histogram(bins = 5, color = "black", fill = "skyblue") +
labs(
  title = "Histogram of Daily Commuting Times",
  x = "Commuting Time (minutes)",
  y = "Frequency"
)
```

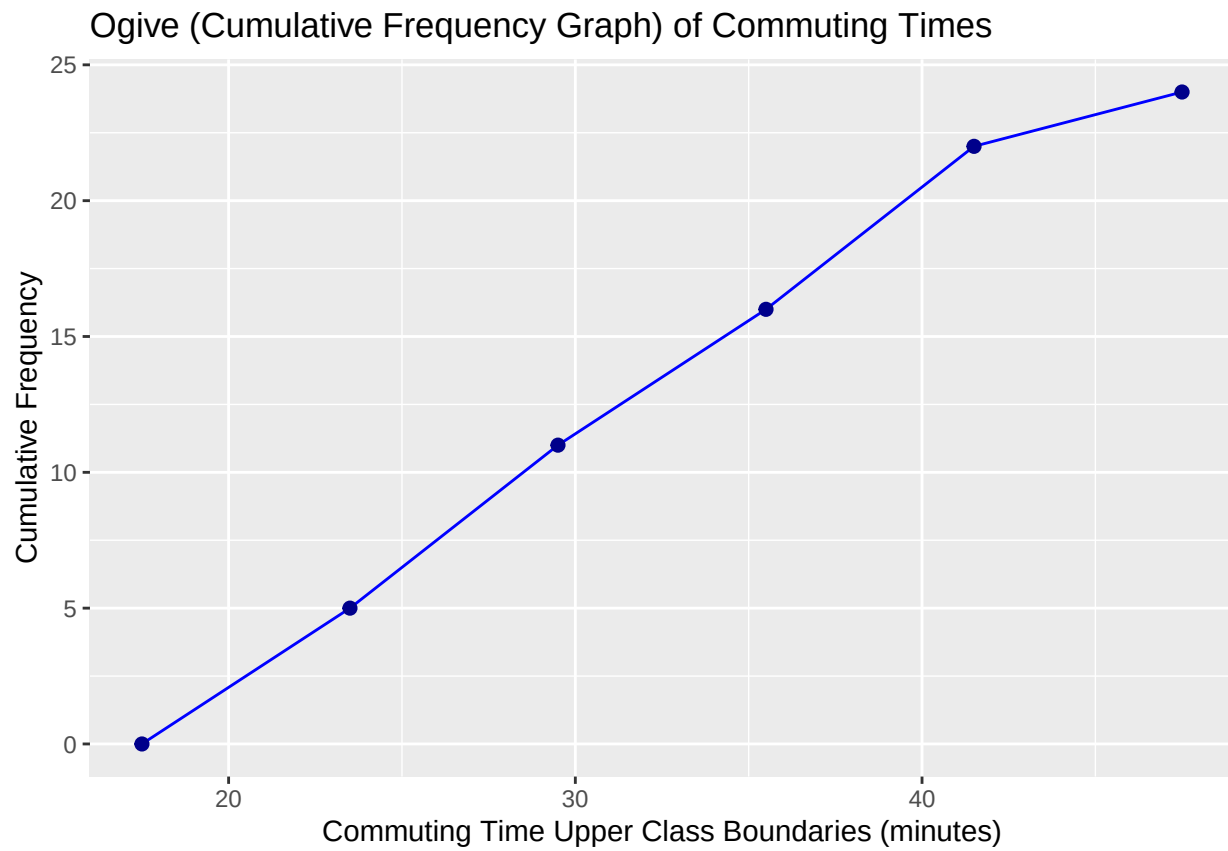


The distribution of commuting times appears relatively symmetric and bell-shaped, centered around the 30 to 35-minute range without noticeable extreme outliers.

Question 3

```
upper_boundaries <- breaks[-1]
ogive_data <- data.frame(
  Boundary = c(breaks[1], upper_boundaries),
  CumFreq = c(0, cum_freq)
)

ggplot(ogive_data, aes(x = Boundary, y = CumFreq)) +
  geom_line(color = "blue") +
  geom_point(color = "darkblue", size = 2) +
  labs(
    title = "Ogive (Cumulative Frequency Graph) of Commuting Times",
    x = "Commuting Time Upper Class Boundaries (minutes)",
    y = "Cumulative Frequency"
  )
```



From the graph and data, approximately 15 to 16 students commute less than 35 minutes.